Shaz (Ethan Shahrooz) Moghaddam

Data Scientist · Python Developer

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Right to work: British citizen · DBS check available on request

PROFILE

Data scientist and Python developer with a Data Science Bootcamp qualification from Imperial College London and HyperionDev (96% average). I build data products end to end, from cleaning raw data and training models to deploying working, tested applications. My most recent is LedgerLet, a full-stack Making Tax Digital platform that files UK landlords' quarterly tax updates to HMRC. Other work includes VoltEdge and VoltScope, two energy data tools combining anomaly detection, AI-powered bill extraction and ESG reporting, and ClinIQ, a clinical trial monitoring dashboard. I have also published seven Android and Flutter apps on Google Play, so I am used to owning a product from first idea to release.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Dart (Flutter), Kotlin, Java, HTML, CSS

Data and ML libraries: pandas, NumPy, SciPy, scikit-learn, TensorFlow, spaCy, NetworkX, Pydantic

Machine learning: linear and logistic regression, decision trees, random forests, gradient boosting, K-means clustering, anomaly detection (Isolation Forest), causal discovery (PC algorithm), NLP

AI tools: Claude, ChatGPT and Gemini, used daily in development and built into products through their APIs for structured data extraction and plain-English questions about data

Frameworks: FastAPI, Flask, Plotly Dash, Streamlit, React, Vite, Node.js, Electron

Databases: PostgreSQL, SQLite, SQLAlchemy 2.0, Alembic migrations

Visualisation: Matplotlib, Seaborn, Plotly, Power BI, Tableau

Cloud and tools: Docker (multi-stage builds), AWS (ECS Fargate), GCP (Cloud Run), Render, Railway, Firebase, Git, GitHub Actions (CI), Jupyter, REST APIs, OAuth 2.0 and OIDC (Auth0)

Testing: pytest, Vitest, Playwright (browser and accessibility tests)

Mobile: Android (Kotlin, Java), Flutter, Google Play Store

SELECTED PROJECTS

LedgerLet, Making Tax Digital Platform for UK Landlords

Python · FastAPI · PostgreSQL · SQLAlchemy · Alembic · React · TypeScript · Auth0 · Docker · GitHub Actions

- Full-stack platform that keeps landlords' property records and files cumulative quarterly updates to HMRC through its Making Tax Digital APIs. Runs end to end on HMRC's sandbox.
- Bank statement CSV import with a two-step preview, column mapping and duplicate detection enforced by row fingerprints in the database. Money is stored as integer pence and mapped to HMRC categories, with joint-ownership shares applied by a tested rounding rule.
- Connects to HMRC over OAuth with encrypted tokens, sends validated fraud prevention headers and keeps an append-only audit trail of every submission. Sign-in through Auth0, hosted in the UK.
- Modular monolith with 394 backend tests on SQLite and PostgreSQL, a React and TypeScript frontend whose types are generated from the API, and a GitHub Actions CI gate with Playwright browser and accessibility tests.

ledgerlet.onrender.com (live demo) · Source code available on request

VoltEdge, Energy Intelligence Platform

FastAPI · Plotly Dash · scikit-learn · NetworkX · SQLite · Docker · Claude API

- Monitors energy use across four industrial sites using 90 days of demo data, with anomaly detection for each site (Isolation Forest) and root-cause analysis through causal discovery (PC algorithm), scored by causal strength.
- ESG reporting covering Scope 2 emissions, GRI 302-1 and carbon intensity, structured for CSRD, plus an AI assistant (Claude) that answers plain-English questions about site data.
- JWT login with role-based access, an audit log of user actions and 903 passing tests. Deployed with Docker on Render.

github.com/ShazMoghaddam/voltedge · Live demo

VoltScope, Energy Bill Extraction and Validation
FastAPI · Claude API · Pydantic · SQLite · Docker

- Reads UK commercial energy bills from PDFs and phone photos (including iPhone HEIC) and extracts them into a fixed Pydantic schema using the Claude API. Also works as the bill module of VoltEdge.
- Rule-based validation checks MPANs, reconciles subtotals and VAT, and flags estimated readings, duplicate charges and contract renewal dates. A portfolio view highlights possible overcharges and compares suppliers.
- Review app with drag-and-drop upload, API-key authentication and JSON, Excel and CSV export. 50 passing tests based on real anonymised bills.

github.com/ShazMoghaddam/voltscope

ClinIQ, Clinical Trial Site Performance Dashboard
FastAPI · Plotly Dash · SQLAlchemy · scikit-learn · spaCy · Claude API · Docker

- Designed for mid-sized CROs in the UK and EU to compare trial sites on enrolment speed, risk and protocol deviations.
- Analyses deviation reports with spaCy, scores site risk with scikit-learn, and includes an AI assistant (Claude) for plain-English questions about trial data.

github.com/ShazMoghaddam/cliniq

SalaryAxis, UK Salary Benchmarking Platform
Python · Flask · React · pandas · NumPy · SciPy · Docker · ONS open data

- Salary benchmarking built on ONS ASHE data, with CPI inflation adjustment, regional and occupation breakdowns, gender pay gap trends and CAGR analysis.
- Python REST API with a React dashboard, packaged with multi-stage Docker builds for AWS ECS Fargate or GCP Cloud Run. No personal data is collected.

github.com/ShazMoghaddam/salaryaxis

World Map, Interactive Choropleth Explorer
HTML · CSS · JavaScript · SVG

- Interactive map of 172 countries across seven data layers, including GDP per capita, HDI, CO_2 emissions and life expectancy, with filters, custom data import, shareable links and embedding. A single HTML file with no dependencies, MIT licensed.

github.com/ShazMoghaddam/world-map · Live demo

EXPERIENCE

Independent Product and Software Developer 2022 – Present
Remote · Google Play Store · Gumroad, Lemon Squeezy and Envato

- Built and published seven Android and Flutter apps on Google Play, owning each one from first idea through to release and updates.
- Sell digital products including finance tools (Market Pulse, TradeHours Pro), generative art (Typography Digits, KNOK Matrix) and screensavers (Aero Atlas), improving them based on customer feedback.
- Published Orbs, an open-source JavaScript library of ten morphing dot animations for loaders and AI “thinking” states: one file of about 5 KB with no dependencies, supporting reduced motion and keyboard use (github.com/ShazMoghaddam/orbs).
- Built the data and AI projects listed above during this period.

Career Break Mar 2013 – 2022
Full-time primary carer for my family.

Sales Representative (Technology) Mar 2011 – Mar 2013
John Lewis and Currys (via agency) · London

- Advised customers on technology products across several London stores, building communication and prioritisation skills.

EDUCATION AND CERTIFICATIONS

Data Science Bootcamp, Imperial College London and HyperionDev June 2025
96% average. Covered SQL and databases, data cleaning and preprocessing, statistics, visualisation (Tableau and Python), supervised learning (linear and logistic regression, decision trees, random forests), K-means clustering and NLP, finishing with three capstone projects.

The Data Science Course: Complete Data Science Bootcamp (Udemy, 32 hours) Oct 2025

iOS and Swift: The Complete iOS App Development Bootcamp (Udemy, 60.5 hours) Jan 2024

100 Days of Code and The Complete Python Bootcamp (Udemy, 58 hours) Oct 2022 – Jul 2023

Google Play Store Listing Certificate (Google Play Academy) Apr 2026

ADDITIONAL

Writing: Seven personal essays on Medium about growth, identity, technology and perspective (medium.com/@shaz.moghaddam).

Availability: Full-time, freelance or contract. On-site, hybrid or remote in London, and open to international opportunities.